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Specialization: *Supervised Learning*

Business Focus: *Real Estate*

Tool: *Pandas, Matplotlib, Seaborn, Sklearn*

Case Study: Predicting House Prices for Maple Valley Realty

Project Learning Opportunities

- *Business Problem Understanding*
- *Data Preprocessing and Cleaning*
- *Understanding Regression Analysis*
- *Model Development and Evaluation*
- *Interpreting Model Results*

Tools and Technology to be Used

jupyter

matplotlib

seaborn

pandas

scikit learn contrib

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Background

- Maple Valley Realty is a regional real estate agency specializing in residential property sales and rentals. They assist buyers, sellers, and investors by offering comprehensive market analyses and property recommendations. The company aims to expand its data-driven approach to improve customer service and streamline operations.

- **Mission:** To empower property buyers and sellers with transparent, data-driven insights to make informed decisions.

- **Vision:** To be the most trusted and innovative real estate agency in the Pacific Northwest.

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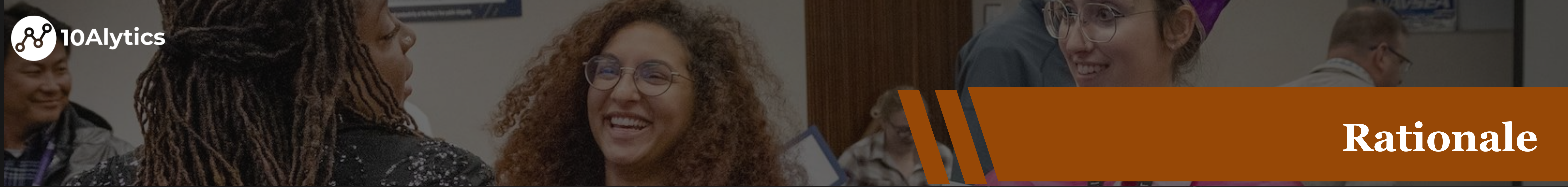
Problem Statement

- Maple Valley Realty wants to develop a predictive model to estimate house prices based on property features and market data.
- Accurate price predictions will help the company:
  - ✓ Advise sellers on optimal pricing strategies.
  - ✓ Provide buyers with fair market value estimates.
  - ✓ Assist investors in identifying undervalued properties.



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Rationale

Real estate prices depend on various factors like location, size, amenities, and market trends. Mispricing properties can lead to lost sales opportunities or dissatisfied clients. By leveraging historical data, the company aims to create a reliable regression model to predict house prices.

This project will:

- **Optimize Pricing**: Ensure properties are priced competitively.
- **Improve Customer Experience**: Enhance trust by providing accurate price estimates.
- **Streamline Operations**: Equip agents with data-driven tools to close deals faster.

Tasks

- **Data Understanding:**
  - ✓ Explore historical property sales data and identify relevant features for price prediction.
  - ✓ Understand potential external factors like neighborhood demographics or economic trends.
- **Data Preparation:**
  - ✓ Clean the dataset, handling issues like missing values, duplicates, and outliers.
  - ✓ Encode categorical variables and scale numerical features.
- **Model Development:**
  - ✓ Build a linear regression model to predict house prices.
- **Model Evaluation:**
  - ✓ Assess model performance using metrics like Mean Absolute Error (MAE) and R^2.
  - ✓ Identify and explain influential factors affecting house prices.
- **Highlight Business Insights:**
  - ✓ Present insights on the key drivers of house prices.

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Data Dictionary

Feature	Description
PropertyID	Unique identifier for each property.
Location	Neighborhood or area where the property is located.
PropertyType	Type of property (e.g., Single Family, Condo, Townhouse).
SizeInSqFt	Total area of the property in square feet.
Bedrooms	Number of bedrooms in the property.
Bathrooms	Number of bathrooms in the property.
YearBuilt	Year when the property was constructed.
GarageSpaces	Number of garage spaces available.
LotSize	Lot size in square feet.
NearbySchools	Average rating of nearby schools (1–10 scale).
MarketTrend	Recent market trend indicator for the area (e.g., 1 for increasing, 0 for stable, -1 for decreasing).
SellingPrice	Actual selling price of the property (Target variable).

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Project Workflow

STEP 1

STEP 2

STEP 3

STEP 4

STEP 5

Data Preparation:

- *Clean the dataset, handling issues like missing values, duplicates, and outliers.*
- *Encode categorical variables and scale numerical features.*

Exploratory Data Analysis (EDA):

- *Perform univariate, bivariate, and multivariate analysis to uncover insights.*
- *Identify relationships between features and attrition.*

Model Development:

- *Build a linear regression model to predict house prices.*

Model Evaluation:

- *Assess model performance using metrics like Mean Absolute Error (MAE) and R^2.*
- *Identify and explain influential factors affecting house prices.*

Present Business Insights:

- *Present insights on the key drivers of house prices.*

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READY TO DELVE IN?